

Mandatory Assignment 1

Development of Large Systems

This assignment is part of the final project described in the final project description document.

Working on the semester project

Form a working group and register your team in the excel file on our OneDrive.

- Come up with your own project based on the final project requirements.
- Define the system requirements:
 - functional
 - non-functional requirements
- Choose a project management strategy for your project.
- Choose your technology stack.

Create a public GitHub repository for your project and provide the link in the excel file on our OneDrive.

System architecture design

- Draw the system architecture - high-level architecture diagram with named microservices and communication styles (REST, GraphQL, messaging).
- Include the frontend application in the system architecture and describe how it interacts with backend services using REST and GraphQL.
- Choose proper communication channels between the microservices and the outside world.
- Define the architectural patterns that you will follow (for example CQRS, Snapshot pattern, etc.).
- Identify the planned AI use case and integration approach.

Architecture design process

The system architecture should be designed collaboratively by the whole team before implementation begins. At this stage, students should:

- Start from the main business workflows and identify the major system responsibilities.
- Propose an initial decomposition into microservices with clearly defined responsibilities and ownership boundaries.

- Discuss and agree on how services will communicate (REST, GraphQL, asynchronous messaging) and what data each service owns.
- Iteratively refine the architecture as a team and document or justify major architectural decisions.
- Identify dependencies and integration points early to support parallel development in later stages.

Divide responsibilities within the team for the implementation of each part of the system.

Deployment considerations

- Briefly discuss how the system could later be deployed in a cloud environment and what architectural implications this may have.
- Identify any expected deployment concerns (e.g., scalability, latency, managed databases, messaging, cost, geographic distribution).
- No final cloud provider decision is required at this stage.
- Describe your CI/CD pipeline.
- Describe your initial testing strategy, including planned unit tests, integration tests, and system-level cooperation tests, and how these will be integrated into the CI/CD pipeline.
- Investigate the options for logging and monitoring of your system in production.

Documentation

- Define documentation strategy.
- Define versioning strategy: databases, APIs, code.

Report

Start writing the final project report (follow the required structure) - concentrate on the topics from this assignment.

Presentation

- Each working group will present their results during the class.
- Discussion
- Feedback - what to improve

Delivery to itsLearning before the deadline

- Provide a link to your group GitHub monorepo
- Upload your PowerPoint presentation
- Report – covering this part

